

DATABRICKS PLATFORM ENABLEMENT · DAY 2

Spark, Workspace &

Lakehouse

The engine, the environment, and the architecture

Go under the hood of Apache Spark, master the Databricks Workspace, and see how the Lakehouse unifies engineering, analytics and AI on one governed platform.

01 Apache Spark
Fundamentals

02 Workspace Components

03 Lakehouse Architecture

Audience: Data Engineers · Analysts · Data Scientists · Platform & Cloud Architects

Builds on Day 1 — Big Data fundamentals & the Lakehouse foundations

What We'll Build Today



PART 01

Apache Spark Fundamentals

- What & why Spark
- Architecture & components
- Execution flow & Spark vs Hadoop
- Use cases & best practices

OUTCOME Explain how Spark runs a job in parallel.



PART 02

Workspace Components

- Workspace & notebooks
- Clusters, jobs & lifecycle
- Repos & Git integration
- Unity Catalog & best practices

OUTCOME Navigate and operate the Databricks Workspace.



PART 03

Lakehouse Architecture

- Evolution & Delta Lake
- Medallion & governance
- AI / ML integration
- End-to-end data flow

OUTCOME Design a governed, end-to-end Lakehouse pipeline.

From the engine, to the environment, to the architecture — one continuous build.

PART ONE

Apache Spark Fundamentals

- What & why Spark · architecture & components
- Execution flow · Spark vs Hadoop
- Use cases & best practices

What Is Apache Spark?

Spark

Workspace

Lakehouse

DEFINITION

Apache Spark is a **unified, open-source, distributed engine** for large-scale data processing — running computations **in memory** across a cluster to power batch, streaming, SQL, machine learning and graph workloads.

- One engine, many workloads — no separate tools to stitch
- Lazy evaluation + a DAG optimizer plan work efficiently
- The compute foundation beneath Databricks & the Lakehouse

100x

faster than disk-based
MapReduce (in-
memory)

4 langs

Python · SQL · Scala ·
R (+ Java) — one API

2009

born at UC Berkeley
AMPLab; open-sourced

1000s

of nodes — scales
from laptop to data
center

Why Spark Won

Spark

Workspace

Lakehouse



In-Memory Speed

Keeps data in RAM and optimizes a DAG of work — up to 100x faster than MapReduce.



Unified Engine

Batch, streaming, SQL, ML and graph share one runtime and one set of APIs.



Easy, High-Level APIs

DataFrames & SQL replace verbose MapReduce code in Python, SQL, Scala or R.



Massive Scalability

Scales out linearly from a single laptop to thousands of cluster nodes.



Fault Tolerance

Lineage (the DAG) lets Spark recompute lost partitions automatically.

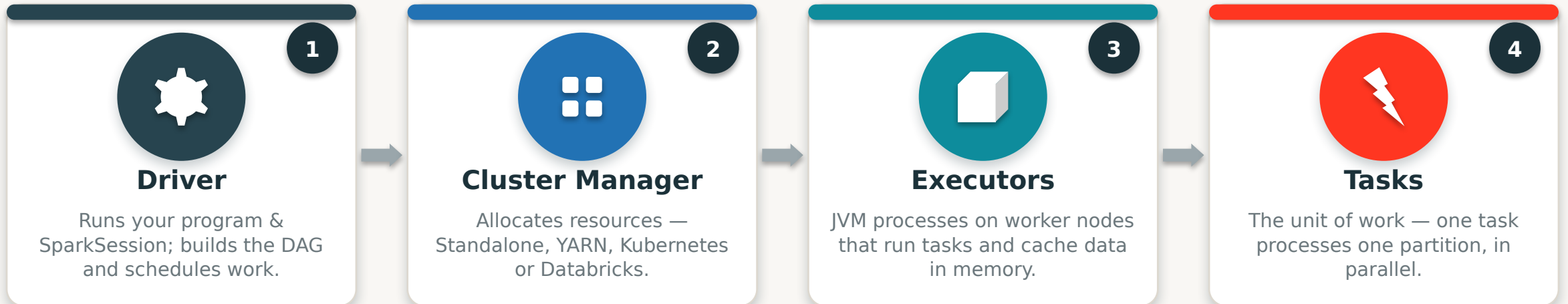


Open Ecosystem

Huge community, rich libraries, and the foundation of the Databricks platform.

Spark Architecture

Spark Workspace Lakehouse



How it runs: the Driver turns your code into a DAG, the Cluster Manager grants Executors, and each Executor runs Tasks in parallel — one per data partition. **On Databricks, all of this is fully managed.**

Master / worker model: one Driver coordinates, many Executors do the work in parallel.

Spark's Unified Component Stack

Spark

Workspace

Lakehouse



1

Spark SQL

ANSI SQL + the Catalyst query optimizer.



2

DataFrames

Typed distributed tables — the core API.



3

Structured Streaming

Real-time pipelines, same DataFrame code.



4

MLlib

Scalable ML algorithms & pipelines.



5

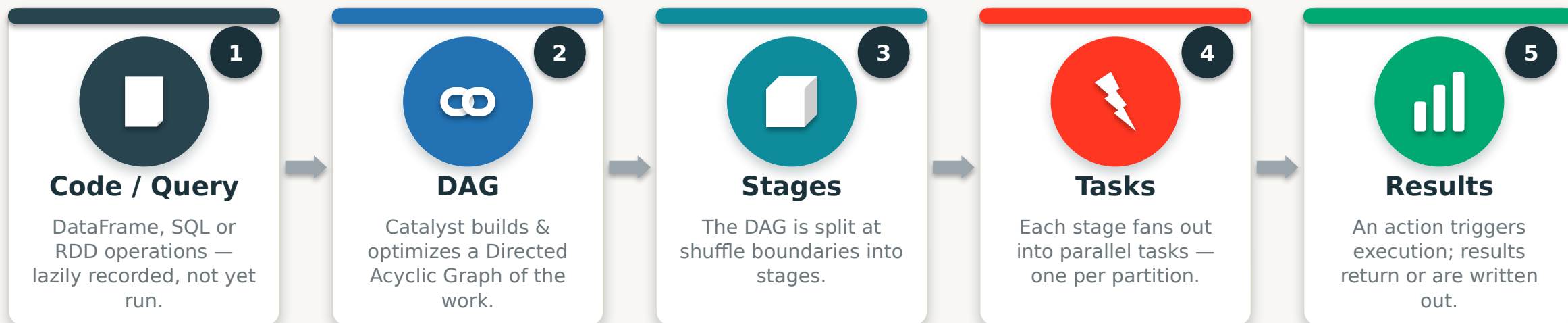
GraphX

Graph processing on connected data.

One engine, one core. Every library sits on the same Spark Core (RDDs, scheduler, memory manager), so you can mix SQL, streaming and ML in a single pipeline — and Photon accelerates them on Databricks.

Spark Execution Flow

Spark Workspace Lakehouse



Lazy by design. Transformations just build the plan; nothing runs until an **action** (e.g. count, write, collect) fires — letting Catalyst optimize the whole DAG first.

Spark vs. Hadoop MapReduce

Spark Workspace Lakehouse

Aspect	Hadoop MapReduce	Apache Spark
Processing	Disk-based, step-by-step	In-memory, DAG-optimized
Speed	Baseline	Up to 100x faster (iterative)
Programming	Verbose low-level Java	Concise DataFrame / SQL APIs
Workloads	Batch only	Batch + streaming + SQL + ML
Real-Time	Not supported	Structured Streaming, near real-time
Iterative / ML	Slow — re-reads from disk	Excellent — caches in memory
Fault Tolerance	Data replication	Lineage (recompute partitions)

What Teams Build with Spark

Spark

Workspace

Lakehouse



Large-Scale ETL

Cleaning, joining and transforming terabytes of data into analytics-ready tables.



Streaming Analytics

Real-time pipelines for fraud, IoT, clickstream and operational dashboards.



Machine Learning

Distributed feature engineering and model training with MLlib & Spark ML.



Interactive SQL & BI

Ad-hoc and production SQL analytics over the data lake at scale.



Graph Analytics

Recommendations, fraud rings and network analysis with GraphX.



Data Lake Processing

The compute behind Delta Lake and the medallion architecture.

Best Practices & Part 1 Takeaways

Spark

Workspace

Lakehouse



Prefer DataFrames / SQL

Let Catalyst & Photon optimize — avoid low-level RDDs unless necessary.



Partition Wisely

Right-size partitions and watch for data skew that overloads one task.



Minimize Shuffles

Shuffles move data across the network — filter early, broadcast small tables.



Cache Reused Data

Persist datasets you reuse across actions to avoid recomputation.



KEY TAKEAWAYS

- Spark = unified, in-memory, distributed engine
- Driver → Cluster Manager → Executors → Tasks
- Lazy: a DAG is optimized, then run by an action
- One core powers SQL, streaming, ML & graph
- Databricks manages & accelerates it (Photon)

PART TWO

Databricks Workspace Components

- Workspace, notebooks & architecture
- Clusters, jobs & Git integration
- Unity Catalog & best practices

The Databricks Workspace

Spark **Workspace** Lakehouse

DEFINITION

The Workspace is the **unified, collaborative environment** where every team builds, runs and governs all data & AI assets — notebooks, clusters, jobs, dashboards, repos and data — in **one browser-based hub**.



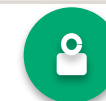
Collaborative

Shared notebooks, comments & co-editing for the whole team.



Browser-Based

Nothing to install — work from anywhere, any cloud.



Governed

Unity Catalog secures every asset with fine-grained control.



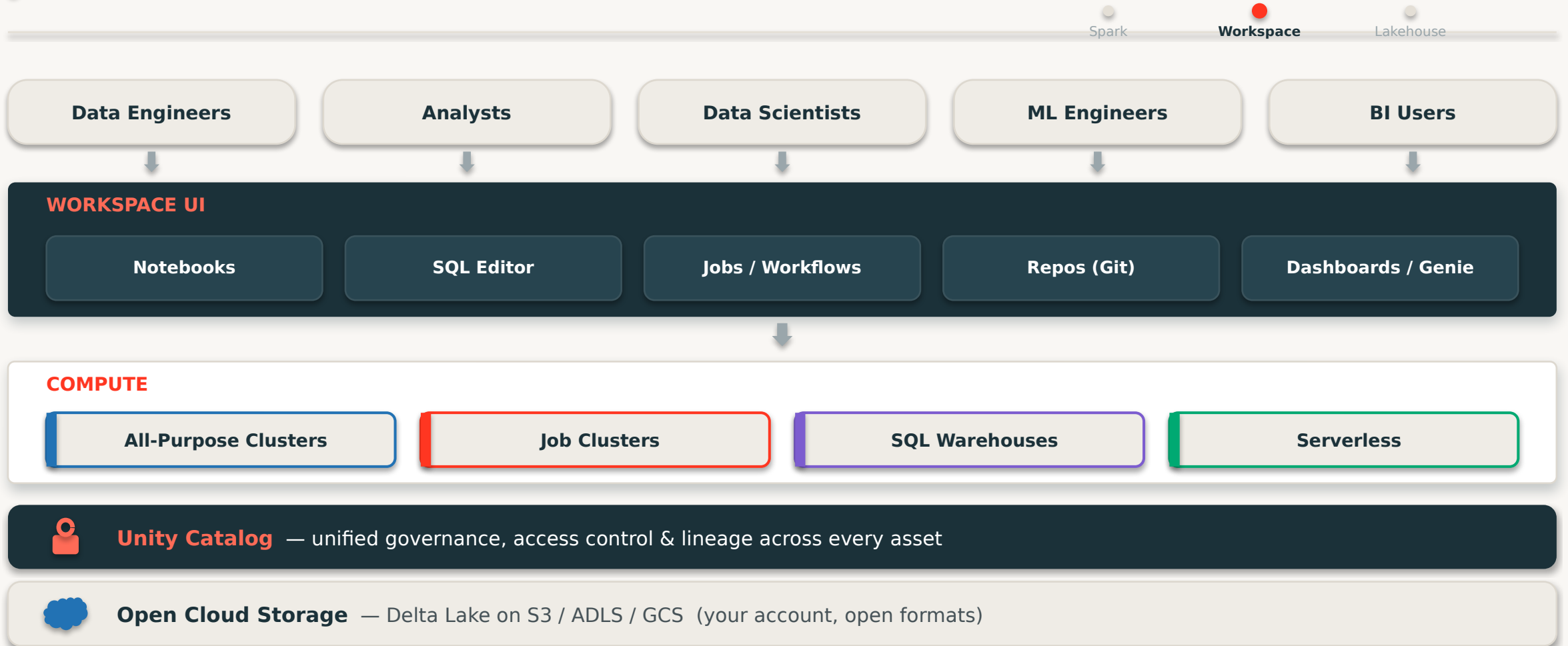
Production-Ready

From exploration to scheduled jobs in the same place.

One place for everyone. Engineers, analysts and scientists share the same notebooks, compute and governed data — eliminating the hand-offs and tool-switching of traditional stacks.

WORKSPACE COMPONENTS

Workspace Architecture



The Seven Building Blocks

Spark

Workspace

Lakehouse



Workspace

The collaborative hub organizing all assets, folders & permissions.



Notebook

Multi-language (Python/SQL/Scala/R) cells for code, viz & narrative.



Cluster

The Spark compute that notebooks and jobs attach to and run on.



Jobs / Workflows

Schedule and orchestrate notebooks & tasks into production pipelines.



Repos

Native Git integration for version control and CI/CD.



Unity Catalog

Central governance for data, ML models, files & permissions.

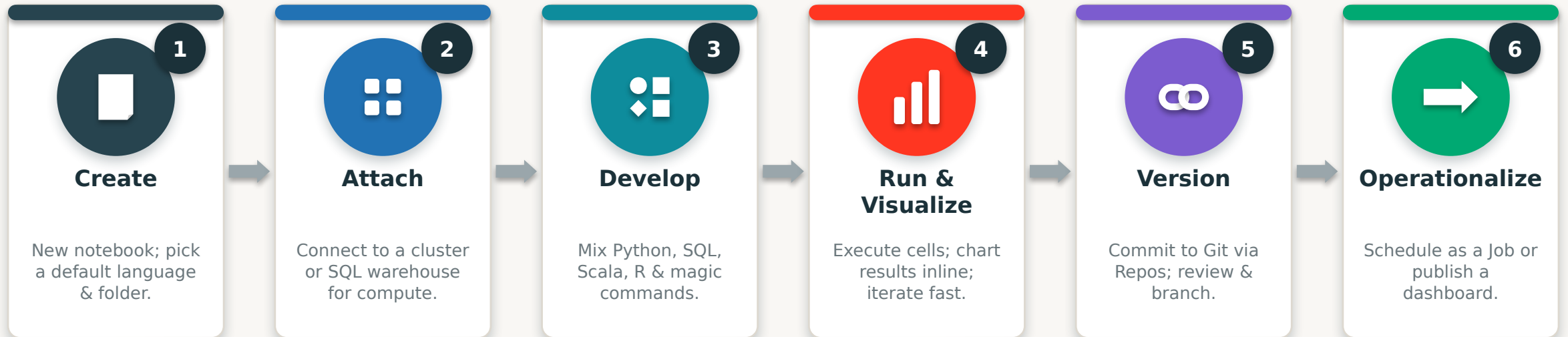


Dashboards

AI/BI dashboards & Genie for SQL analytics and natural-language Q&A.

The Notebook Lifecycle

Spark **Workspace** Lakehouse



Exploration to production, in one artifact. The same notebook you prototype in can be version-controlled and scheduled — no rewrite, no hand-off to a different tool.

Cluster Types — Choosing Compute

Spark **Workspace** Lakehouse

Aspect	All-Purpose Cluster	Job Cluster
Purpose	Interactive development	Automated production runs
Lifecycle	Long-running, manual start	Created per job, auto-terminated
Sharing	Shared by a team	Dedicated to one job
Cost	Higher — stays on	Lower — runs only when needed



SQL Warehouse

Photon-optimized compute for BI & SQL analytics

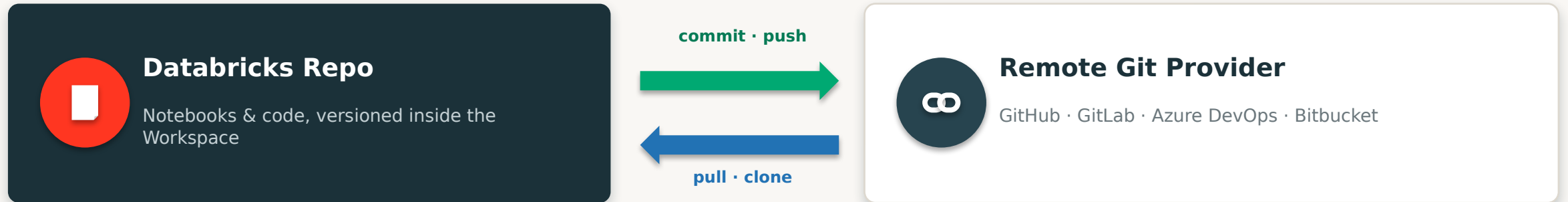


Serverless

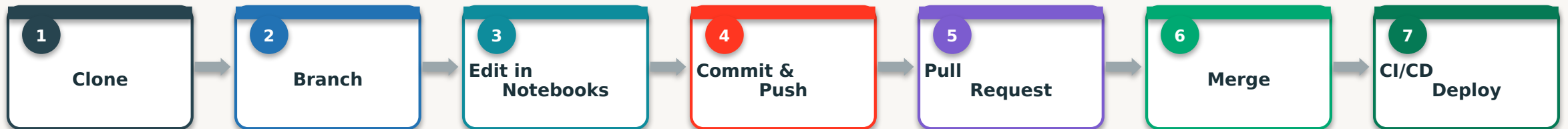
Instant, fully-managed pools — zero infra to configure

Repos & Git Integration

Spark **Workspace** Lakehouse



TYPICAL WORKFLOW



Engineering discipline, built in. Repos bring branches, pull requests and CI/CD to notebooks — so data pipelines are reviewed, tested and deployed like any other software.

Unity Catalog — Unified Governance

Spark

Workspace

Lakehouse

ONE NAMESPACE FOR EVERY ASSET

Catalog *top-level container*

Schema *database / namespace*

Table · View · Volume · Model *the governed asset*

catalog . schema . object

One name & permission across all clouds & workspaces



Fine-Grained Access

Row/column security via standard SQL GRANTS tied to SSO identity.



Automated Lineage

Column-level lineage & audit across notebooks, jobs & dashboards.



Discovery & Sharing

Search, tags and open Delta Sharing across organizations.



Data + AI Assets

Governs tables, files, ML models & functions in one place.

Best Practices & Part 2 Takeaways

Spark

Workspace

Lakehouse



Version Everything

Use Repos & Git for all notebooks — branches, reviews, CI/CD.



Job Clusters for Prod

Run scheduled pipelines on ephemeral job clusters to cut cost.



Parametrize Notebooks

Use widgets & job parameters to make code reusable and testable.



Govern with Unity Catalog

Centralize permissions; grant least privilege, audit lineage.



KEY TAKEAWAYS

- Workspace = one collaborative, governed hub
- Notebooks span exploration → production
- All-purpose for dev, job clusters for prod
- Repos bring Git & CI/CD to data work
- Unity Catalog governs every data & AI asset

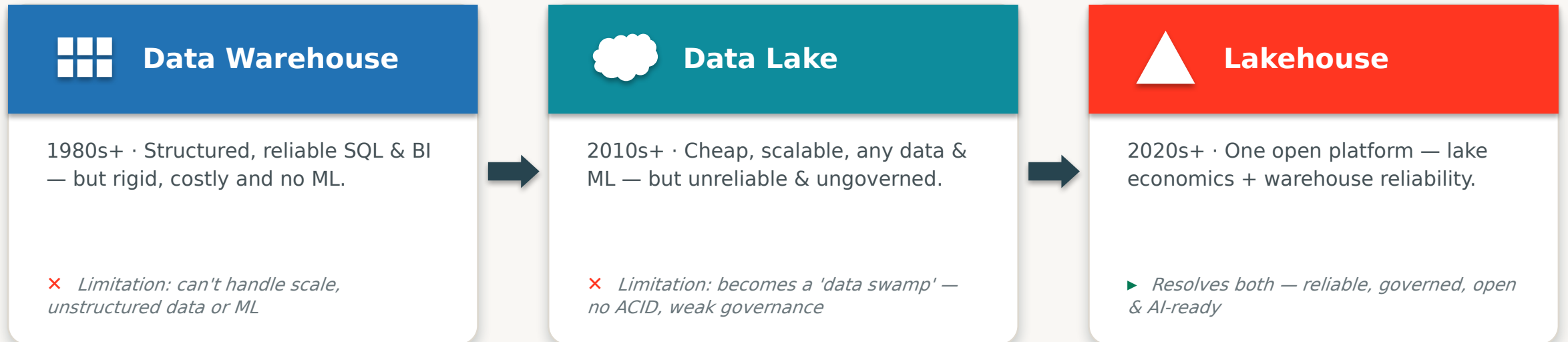
PART THREE

Databricks Lakehouse Architecture

- Evolution · Lakehouse & Delta Lake
- Medallion · governance · AI & ML
- End-to-end pipeline & benefits

Evolution of Data Platforms

Spark Workspace **Lakehouse**



Each generation fixed the last one's flaw. The Lakehouse keeps the lake's low-cost openness and adds the warehouse's reliability and performance — so you no longer need both.

What Is the Lakehouse?

Spark

Workspace

Lakehouse

DEFINITION

A Lakehouse is a single platform that adds a **transactional, governed layer (Delta Lake)** directly on low-cost cloud object storage — delivering **warehouse reliability & performance with lake openness & scale** for every workload.

- One open copy of data — no warehouse/lake duplication
- ACID transactions & performance on cheap storage
- Serves BI, SQL, streaming, data science & AI together

Open

Delta & Parquet — no proprietary lock-in

Reliable

ACID transactions & schema enforcement

Unified

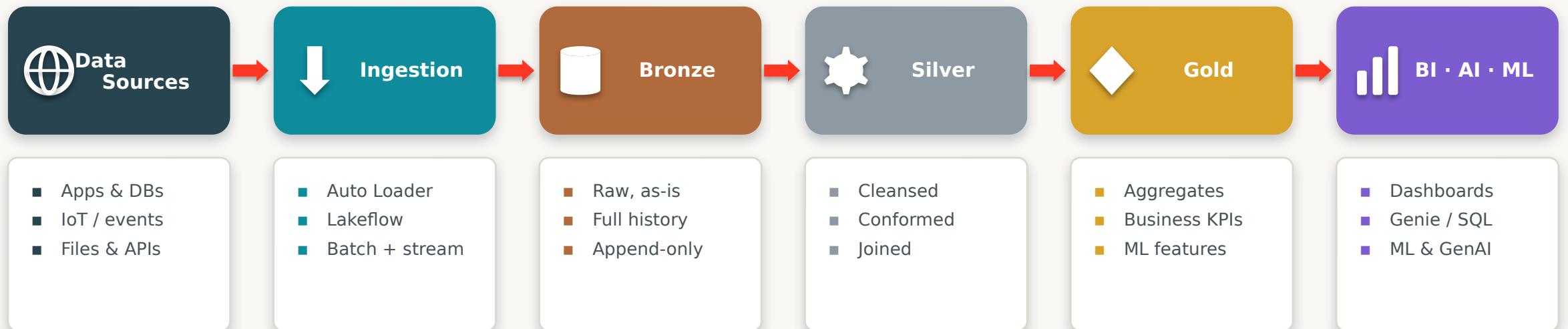
all workloads on one copy of data

Governed

one security model via Unity Catalog

Lakehouse Architecture — Layer by Layer

Spark Workspace **Lakehouse**



Delta Lake open storage for every layer **Unity Catalog** governs the entire pipeline

Raw data flows left-to-right through Delta tables of rising quality — all open, all governed.

The Medallion Architecture — In Detail

Spark

Workspace

Lakehouse



Delta Lake Fundamentals

Spark

Workspace

Lakehouse



1

ACID Transactions

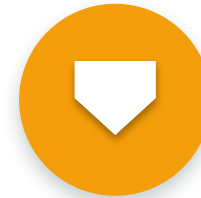
Reliable concurrent reads & writes — no corrupt or partial data.



2

Time Travel

Query or roll back to any earlier version of the table.



3

Schema Enforcement

Rejects bad data on write; supports safe schema evolution.



4

Data Versioning

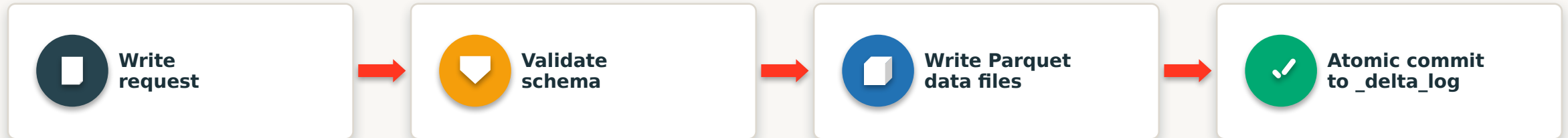
Every commit is a version — full audit history & reproducibility.

Under the hood: a Delta table is just **Parquet data files + a JSON transaction log (`_delta_log`)** — the log is what delivers all four guarantees, plus MERGE, OPTIMIZE & Z-ORDER.

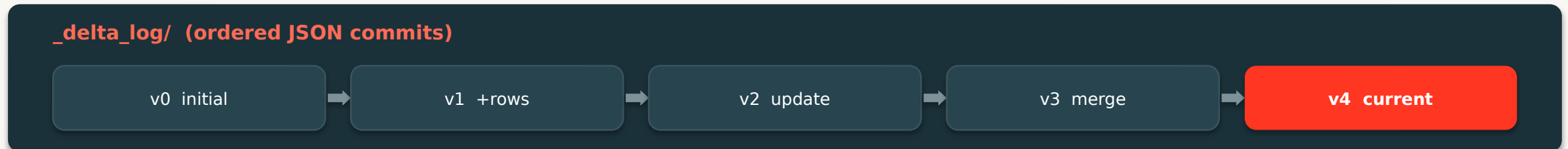
Delta Lake — Transaction Flow

Spark Workspace Lakehouse

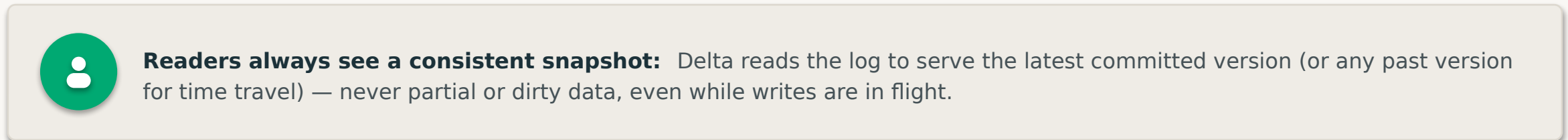
WRITE PATH



`_delta_log/` (ordered JSON commits)



READ PATH



Governance & Open Sharing

Spark

Workspace

Lakehouse



Centralized Access

One permission model — ANSI SQL GRANTS with row & column security.



Lineage & Audit

Automatic column-level lineage and full audit logs for compliance.



Delta Sharing

Open protocol to share live data across orgs — no copying, any client.



Discovery & Tags

Search, tag and classify all assets in one governed catalog.



Compliance Ready

Encryption, network isolation and GDPR / HIPAA / SOC 2 controls.

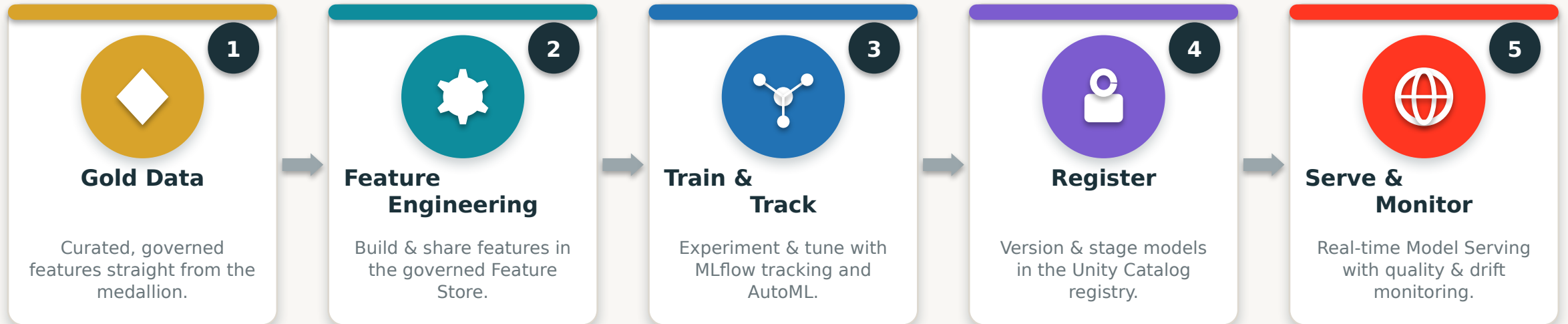


Data + AI Together

Govern tables, files, ML models & features under one roof.

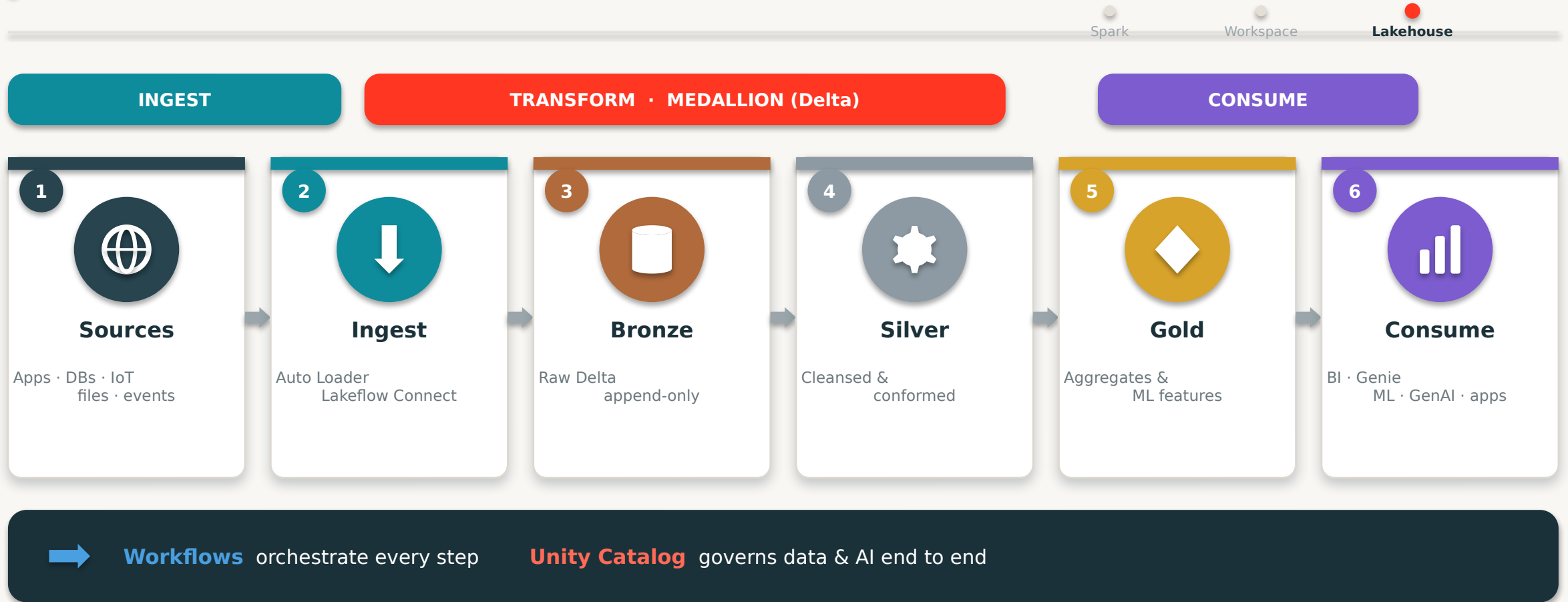
AI & ML on the Lakehouse

Spark Workspace **Lakehouse**



Mosaic AI: the full ML & GenAI lifecycle on governed data — MLflow, AutoML, Feature Store, Model Serving, Vector Search and LLM fine-tuning, all under Unity Catalog.

End-to-End Data Pipeline



Spark + Photon execute every stage · Delta stores it · Workflows run it · Unity Catalog governs it.

Benefits of the Lakehouse

Spark

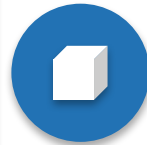
Workspace

Lakehouse



Unified

One platform for ETL, BI, streaming, data science & AI.



Open

Delta & Parquet open formats on your storage — zero lock-in.



Reliable

ACID transactions and schema enforcement on the lake.



Performant

Photon and Delta optimizations deliver warehouse-class speed.



Governed

One security & lineage model for all data and AI assets.



Cost-Effective

One copy of data, elastic compute — no duplicate systems.

Real-World Example — Omni-Channel Retailer

Spark

Workspace

Lakehouse

THE PIPELINE



THE OUTCOMES

10x

faster pipelines vs the legacy warehouse + lake

Real-time

personalization & inventory visibility

1 copy

of governed data for BI, ML and GenAI

~50%

lower total cost of ownership

Day 2 — Key Takeaways

1 Apache Spark

- Unified, in-memory, distributed engine
- Driver → Cluster Mgr → Executors → Tasks
- Lazy DAG; one core for SQL/stream/ML



Spark is the engine; Databricks makes it effortless and fast (Photon).



Delta Lake + medallion turn raw files into reliable, governed tables.

2 Workspace

- One collaborative, governed hub
- Notebooks: exploration → production
- Repos + Git; Unity Catalog governance



The Workspace unifies notebooks, compute, Git and governance in one hub.



One platform, one governed copy of data — for analytics and AI alike.

3 Lakehouse

- Delta + medallion = trusted data
- ACID, time travel on open storage
- Governed BI, ML & GenAI, end to end

Thank You

Day 2 complete — you can now read the engine, the workspace and the Lakehouse architecture end to end.

KEEP BUILDING

docs.databricks.com

[Databricks Academy](#)

[Free Edition — hands-on](#)

[Delta Lake & Spark docs](#)



Spark



Workspace



Clusters



Delta



Medallion



Unity Catalog



Next: hands-on lab & Delta Live Tables